Further Integration Techniques

ANSWER KEY

Integration by substitution

1 Evaluate $\int 2x(x^2+1)^4\,dx$ using the substitution $u = x^2 + 1$.
$du = 2x\,dx$. So $\int (x^2+1)^4 \cdot 2x\,dx = \int u^4\,du = \frac{u^5}{5} + C = \frac{(x^2+1)^5}{5} + C$.

2 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x\,dx$ using the substitution $u = \sin x$.
$du = \cos x\,dx$. When $x = 0$, $u = 0$; when $x = \frac{\pi}{2}$, $u = 1$. $\int_0^1 u^3\,du = \left[\frac{u^4}{4}\right]_0^1 = \frac{1}{4}$.

Integration by parts

3 Evaluate $\int xe^x\,dx$ using integration by parts.
Let $u = x$, $dv = e^x\,dx$, so $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C = (x-1)e^x + C$.

4 Evaluate $\int x \ln x\,dx$ using integration by parts.
Let $u = \ln x$, $dv = x\,dx$, so $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$. $\int x\ln x\,dx = \frac{x^2}{2}\ln x - \int \frac{x^2}{2}\cdot\frac{1}{x}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

Repeated integration by parts

5 Evaluate $\int x^2 e^x\,dx$, applying integration by parts twice.
First application: $u = x^2$, $dv = e^x dx$: $\int x^2 e^x\,dx = x^2 e^x - \int 2xe^x\,dx$. From the previous problem, $\int xe^x\,dx = (x-1)e^x + C$, so $\int 2xe^x\,dx = 2(x-1)e^x$. Result:
$x^2e^x - 2(x-1)e^x + C = (x^2 - 2x + 2)e^x + C$.

Integration using trigonometric identities

6 Evaluate $\int \cos^2 x\,dx$ using the identity $\cos^2 x = \frac{1+\cos 2x}{2}$.
$\int \frac{1+\cos 2x}{2}\,dx = \frac{1}{2}\left(x + \frac{\sin 2x}{2}\right) + C = \frac{x}{2} + \frac{\sin 2x}{4} + C$.

Integration using partial fractions

7 Evaluate $\int \frac{1}{x^2-1}\,dx$ by first expressing the integrand in partial fractions.
$\frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$. Solving: $A = \frac{1}{2}$, $B = -\frac{1}{2}$.
$\int\left(\frac{1/2}{x-1} - \frac{1/2}{x+1}\right)dx = \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C = \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$.

Evaluating a definite integral

8 Evaluate $\int_1^e \frac{\ln x}{x}\,dx$, using the substitution $u = \ln x$.
$du = \frac{1}{x}dx$. When $x = 1$, $u = 0$; when $x = e$, $u = 1$. $\int_0^1 u\,du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}$.

9 Evaluate $\int_0^{\pi} x\sin x\,dx$ using integration by parts.
Let $u = x$, $dv = \sin x\,dx$, so $du = dx$, $v = -\cos x$.
$\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x$. Evaluating from 0 to π:
$[-\pi\cos\pi + \sin\pi] - [0 + 0] = -\pi(-1) + 0 = \pi$.

Integration involving inverse trigonometric results

10 Evaluate $\int \frac{1}{1+x^2}\,dx$, and use it to evaluate $\int_0^1 \frac{1}{1+x^2}\,dx$.
$\int \frac{1}{1+x^2}\,dx = \arctan x + C$. So $\int_0^1 \frac{1}{1+x^2}\,dx = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}$.

Integration by substitution with trigonometric functions

11 Evaluate $\int \sin^3 x\,dx$, using the identity $\sin^2 x = 1 - \cos^2 x$ and the substitution $u = \cos x$.
Rewrite: $\sin^3 x = \sin x(1 - \cos^2 x)$. Let $u = \cos x$, $du = -\sin x\,dx$.
$\int \sin x(1 - \cos^2 x)\,dx = -\int (1 - u^2)\,du = -u + \frac{u^3}{3} + C = -\cos x + \frac{\cos^3 x}{3} + C$.

12 Evaluate $\int \tan x\,dx$, using the substitution $u = \cos x$.
$\tan x = \frac{\sin x}{\cos x}$. Let $u = \cos x$, $du = -\sin x\,dx$.
$\int \frac{\sin x}{\cos x}\,dx = -\int \frac{1}{u}\,du = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$.

Integration by parts with a logarithmic integrand

13 Evaluate $\int_1^e \ln x\,dx$ using integration by parts.
Let $u = \ln x$, $dv = dx$, so $du = \frac{1}{x}dx$, $v = x$. $\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C$. Evaluating from 1 to e: $[e\ln e - e] - [1\ln 1 - 1] = (e - e) - (0 - 1) = 0 + 1 = 1$.

Integration using a Weierstrass-style substitution

14 Evaluate $\int \frac{1}{5 + 4\cos x}\,dx$ using the substitution $t = \tan\left(\frac{x}{2}\right)$, given that this substitution transforms $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2}{1+t^2}dt$, leading to the simplified integrand $\frac{2}{9+t^2}$.
Substituting: $\int \frac{1}{5 + \frac{4(1-t^2)}{1+t^2}} \cdot \frac{2}{1+t^2}\,dt = \int \frac{2}{5(1+t^2) + 4(1-t^2)}\,dt = \int \frac{2}{9+t^2}\,dt$. Using the standard result $\int \frac{1}{a^2+t^2}\,dt = \frac{1}{a}\arctan\left(\frac{t}{a}\right) + C$ with $a = 3$: $2\cdot\frac{1}{3}\arctan\left(\frac{t}{3}\right) + C = \frac{2}{3}\arctan\left(\frac{1}{3}\tan\frac{x}{2}\right) + C$.

Integration by parts with a trigonometric-exponential product

15 Evaluate $\int e^x \cos x\, dx$, applying integration by parts twice and solving the resulting equation for the original integral.

Let $I = \int e^x \cos x\, dx$. First application: $u = \cos x$, $dv = e^x dx$, so $du = -\sin x\, dx$, $v = e^x$: $I = e^x \cos x + \int e^x \sin x\, dx$. Second application on $\int e^x \sin x\, dx$: $u = \sin x$, $dv = e^x dx$: $\int e^x \sin x\, dx = e^x \sin x - \int e^x \cos x\, dx = e^x \sin x - I$. Substituting back: $I = e^x \cos x + e^x \sin x - I$, so $2I = e^x(\cos x + \sin x)$, giving $I = \dfrac{e^x(\cos x + \sin x)}{2} + C$.

Finding the area between two curves using integration

16 Find the area of the region enclosed between $y = \sin x$ and $y = \cos x$ for $\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$ (the interval on which $\sin x \geq \cos x$).

Area $= \int^{5\pi/4}(\sin x - \cos x)\, dx = [-\cos x - \sin x]_{\pi/4}^{5\pi/4}$. At $x = \frac{5\pi}{4}$: $-\cos\frac{5\pi}{4} - \sin\frac{5\pi}{4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$. At $x = \frac{\pi}{4}$: $-\cos\frac{\pi}{4} - \sin\frac{\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}$. Area $= \sqrt{2} - (-\sqrt{2}) = 2\sqrt{2}$.

17 Evaluate $\int x^2 \ln x\, dx$ using integration by parts.

Let $u = \ln x$, $dv = x^2 dx$, so $du = \frac{1}{x}dx$, $v = \frac{x^3}{3}$. $\int x^2 \ln x\, dx = \frac{x^3}{3}\ln x - \int \frac{x^3}{3} \cdot \frac{1}{x}\, dx = \frac{x^3}{3}\ln x - \int \frac{x^2}{3}\, dx = \frac{x^3}{3}\ln x - \frac{x^3}{9} + C$.

18 A particle's velocity is given by $v(t) = t\cos t$ metres per second. Find the particle's displacement over the interval $0 \leq t \leq \pi$, using integration by parts.

Displacement $= \int_0^\pi t\cos t\, dt$. Using integration by parts with $u = t$, $dv = \cos t\, dt$, so $du = dt$, $v = \sin t$: $\int t\cos t\, dt = t\sin t - \int \sin t\, dt = t\sin t + \cos t + C$. Evaluating from 0 to π: $[\pi \sin\pi + \cos\pi] - [0\sin 0 + \cos 0] = [0 - 1] - [0 + 1] = -1 - 1 = -2$. Displacement $= -2$ metres.

19 This question has three parts. (a) Use the substitution $u = x^2 + 4$ to find $\int \frac{x}{x^2 + 4}\, dx$. (b) Use integration by parts to find $\int \arctan x\, dx$. (c) Hence, using the result of part (b) and the standard result $\int \frac{1}{1 + x^2} dx = \arctan x + C$, evaluate $\int_0^1 \arctan x\, dx$, giving an exact answer in terms of π and $\ln$.

(a) $u = x^2 + 4$, $du = 2x\, dx$: $\int \frac{x}{x^2 + 4}\, dx = \frac{1}{2}\int \frac{1}{u}\, du = \frac{1}{2}\ln|x^2 + 4| + C$. (b) Let $u = \arctan x$, $dv = dx$, so $du = \frac{1}{1 + x^2}dx$, $v = x$: $\int \arctan x\, dx = x\arctan x - \int \frac{x}{1 + x^2}\, dx = x\arctan x - \frac{1}{2}\ln(1 + x^2) + C$ (using the result from part (a), with 4 replaced by 1). (c) Evaluating from 0 to 1: $\left[1 \cdot \arctan 1 - \frac{1}{2}\ln 2\right] - \left[0 - \frac{1}{2}\ln 1\right] = \frac{\pi}{4} - \frac{1}{2}\ln 2$.

Reduction formulas

20 Given the reduction formula $I_n = \int x^n e^x\, dx = x^n e^x - nI_{n-1}$ (obtained via integration by parts), and that $I_0 = e^x + C$, use it to find $I_2 = \int x^2 e^x\, dx$.

$I_1 = x^1 e^x - 1 \cdot I_0 = xe^x - e^x$ (ignoring the constant until the final step).
$I_2 = x^2 e^x - 2I_1 = x^2 e^x - 2(xe^x - e^x) = x^2 e^x - 2xe^x + 2e^x + C = (x^2 - 2x + 2)e^x + C$, matching the result found earlier in this worksheet by direct repeated integration by parts.